



Week 3: Executable Authorization Evidence

From UQ-local architecture to a tested synthetic security slice

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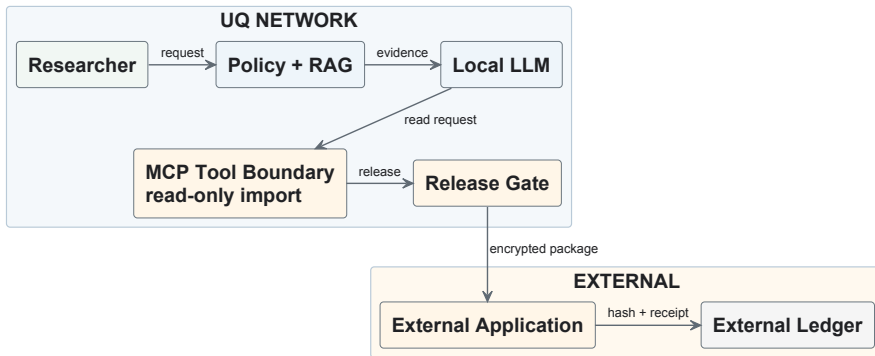
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Target Workflow

Target design: authorization precedes Agent use; the gateway controls external release.





This Week: One Executable Security Slice

I implemented and tested the order: authorize first, retrieve second.

IMPLEMENTED

Synthetic identities and sources

Server-side allow / deny / share / revoke

SQLite forced scope + source reauthorization

Signed evidence bundle + content-free audit chain

26 passing E2 tests

NEXT INTEGRATION

UQ identity and group scopes

PostgreSQL forced RLS + vector retrieval

Stage 1A model gateway and MCP

Tidy3D, Bunya, and secure release

LOCAL LOOPBACK

SYNTHETIC DATA

NO MODEL IN ACCESS DECISION



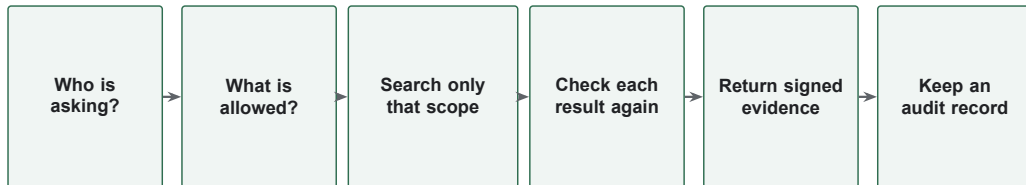
Enterprise E2: Identity-Aware Retrieval

The simple question is: who is asking, and which evidence may this person see?

IMPLEMENTED

SYNTHETIC

MODEL-FREE





Executed Demo I: Tenant Isolation

IL

Industrial Local Agent
Enterprise E2 Security Console

● Logback | No model

SYNTHETIC AUTHORIZATION WORKSPACE

Access control demonstration

01 Identity scenario
Presenter-selected synthetic subject

Alice
tenant-a / researcher

Bob
tenant-b / reviewer

Carol
tenant-b / researcher

02 Retrieval
Query inside the authorized source set

Query preset
Control result

Run retrieval

03 Policy
Explicit same-tenant source access

Share source-a2

Revoke

DECISION STREAM

Observed outcomes

Reset

NO EVIDENCE

Authorized retrieval completed

subject=user-alice tenant=tenant-a policy=1 sources=none bundle=14a3628593f9...

ALLOW

Authorized retrieval completed

subject=user-carol tenant=tenant-b policy=1 sources=source-b1 bundle=92855267d9cd...



Executed Demo II: Share and Revoke

IL

Industrial Local Agent
Enterprise E2 Security Console

● Logback | No model

SYNTHETIC AUTHORIZATION WORKSPACE

Access control demonstration

01 Identity scenario
Presenter-selected synthetic subject

Alice
tenant-a / researcher

Bob
tenant-b / reviewer

Carol
tenant-b / researcher

02 Retrieval
Query inside the authorized source set

Query preset
Ring linewidth

Run retrieval

03 Policy
Explicit same-tenant source access

Share source-a2

Revoke

DECISION STREAM

Observed outcomes

Reset

NO EVIDENCE

Authorized retrieval completed

subject=user-alice tenant=tenant-a policy=3 sources=none bundle=6928765df717...

DENY

Alice's source-a2 share revoked

subject=user-alice policy=3

ALLOW

Authorized retrieval completed

subject=user-alice tenant=tenant-a policy=2 sources=source-a2 bundle=2c0bed308535...

ALLOW

source-a2 shared with Alice

subject=user-alice policy=2



What the Demo Proves

The access contract is testable; the production deployment is the next engineering step.

TENANT ISOLATION

Carol: ALLOW
Alice: NO EVIDENCE

POLICY LIFECYCLE

share → ALLOW → revoke

NEGATIVE PATHS

forged claim, tampered bundle: DENY

AUDIT CHAIN

PASS: signed bundle and content-free hash chain

26 PASSED

CLI + WEB SHARE ONE RUNTIME

SYNTHETIC ONLY

Next integration: UQ identity, production RLS/RAG, model gateway, and approved compute path.



Bunya Model Plan: Quality vs Cost

The LLM is replaceable; Agent contracts stay model-independent.

LOCAL DEMO

3B model

Observed on 12 GB GPU
E2 itself is CPU-only
Lowest cost and fastest iteration

FIRST BUNYA TEST

14B, 4-bit

Roughly 9–12 GB VRAM*
Start with 1 GPU, 8
CPU cores, 32 GB RAM
RCC confirms GPU,
quota, and charges

ESCALATE ONLY IF NEEDED

32B / 70B-class

Roughly 20–24 /
40–48 GB VRAM*
More GPU memory, queue
time, and serving cost
Benchmark must justify it

Cost drivers: GPU allocation, job time, storage, and serving complexity.

*Planning estimates only; actual VRAM and RCC pricing require a Bunya benchmark.